

UNIT 1: SET THEORY



Introduction

Welcome to the first unit in our module. You probably have used the word “Set” from time to time in different situations. We speak about ‘sofa sets’, ‘dining sets’ and so on. In general, we use the word to classify certain things, objects or items that have some common properties. The same applies to mathematics. Sets were first used in mathematics by George Cantor, a German mathematician. He is known to be the father of Set Theory. Set Theory has become basic to most concepts in mathematics and is very useful in clarifying and classifying mathematical ideas, thus making them easier to understand.

In this Unit, we will define a set and discuss various ways of denoting a set. We will also look at different types of sets and finally look at set operations and set relations.



Objectives

By the end of this unit, you must be able to:

- define a set
- describe various ways of presenting, representing and naming a set
- name different types of sets
- use set operations and relations to solve mathematical problems



Key terms

Make sure that you understand the following terms that are going to be used in this unit: Set, element, cardinality, universal set, empty set, subset, complement of a set, union of sets, intersection of sets, and power set.

Reflection Activity: Take some time (about 5 minutes) to reflect on your own understanding of the word ‘Set’. In your own words, describe a set? What are some of the examples of sets that you have in your mind? List down all the things that you can think about that use the word set to describe.



Lesson 1: Set and set notation

A set is any collection of well-defined objects. For instance:

- A coffee table with its 4 stools is known as a coffee set.

- We can also form a set of positive even numbers that are less than 12.

Roster notation

Sets can be represented in many ways, such as listing all of its members between 'curly' brackets $\{ \}$ with elements/members separated by commas. We call this representation of a set the **roster notation**.

Thus, the set of positive even numbers less than 12 can be represented in a roster notation as $\{2, 4, 6, 8, 10\}$

Definition: Elements of a set

The objects that make up the set are called **elements** or **members** of the set.

We shall denote sets by capital letters A, B, C, X, Y, Z etc. while elements shall be denoted by their lower case letters a, b, c, x, y, z etc.

Thus if A is a set and x is a member of A, we write $x \in A$ (read as 'x is a member of set A').

If, on the other hand, x is not a member of A, we write $x \notin A$ (read as 'x is not a member of set A').

Example 1.1

Let V be the set of all vowels. Then $V = \{a, e, i, o, u\}$. Clearly $i \in V$ but $d \notin V$ as d is a consonant.

Example 1.2

Let M be the set of all multiples of 5. Then $35 \in M$ while $21 \notin M$

Example 1.3

The set of positive odd integers greater than 21 may be written as $\{23, 25, 27, 29, \dots\}$, where the dots indicate that the pattern established by the first four entries continues indefinitely (without stopping).

Set builder notation: We can sometimes present sets by describing their common property rather than listing all their elements. This way of representing sets is called **set-builder notation**.

Example 1.4

Use the set builder notation to write down the following sets:

- a) The set, P, of all prime numbers.
- b) The set, M, of the months in a year.

Solution

- a) Let P be the set of prime numbers, then we have $P = \{p : p \text{ is a prime number}\}$, where the colon (sometimes a vertical bar is used) is read as “such that”.
- b) If M is the set of months of the year, instead of listing all the months of the year, using

the set notation we write $M = \{m \mid m \text{ is a month of the year}\}$

Cardinal number of a set

Let A be a set, then the cardinal number of the set (or the **cardinality of the set**) is the number of elements in the set. This is denoted as $n(A)$ or $|A|$.

For instance the set of vowels $V = \{a, e, i, o, u\}$ has 5 elements and so $n(V) = 5$ while

$n(M) = 12$ where $M = \{m \mid m \text{ is a month of the year}\}$

Notice that in the two examples above, we can count the number of elements in a set. Such sets are called **Finite sets** while a set in which there is no end in counting the elements is referred to as an **Infinite set**. A set with exactly one element is called a **Singleton** e.g. the set of even prime numbers is Singleton whose only element is 2.

Example 1.5

The set M of all people that voted in the 2020 presidential election in Malawi is a finite set. So is the set P of political parties that competed in the election.

Example 1.6

Let $P = \{x : x \text{ is an odd positive whole number}\}$ is certainly an infinite set and $n(P) = \infty$.

Note that: ∞ (read as *infinity*) is not a number! It just shows that there is no end in counting)



Activity 1a

1. Say whether the following statements are True or False
 - a) $5 \in \{3, 7, 5, 47, 8, a\}$
 - b) $7 \in \{x : x \text{ is an even number}\}$
2. Write the following sets using the set builder notation
 - a) the set of odd whole numbers greater than 0 and less than 10
 - b) the set D of the days of the week
 - c) the set of odd whole numbers
3. Find the cardinality of the sets in question 2(a), 2(b) and 2(c)
4. If $C = \{\text{Blantyre}, \{\text{Chitipa}, \text{Karonga}, \text{Nkhata-Bay}\}, \{\text{Dowa}, \text{Salima}\}\}$, find $n(C)$.



Lesson 2: Relation between Sets

Equality of sets

Definition: Let A and B be sets. Therefore, $A = B$ if every member of A is also a member of B and every member of B is also a member of A. In other words, sets can only be equal if they have exactly same elements. If this is not the case, then A is not equal to B, written $A \neq B$.

Example 1.7

Let $A = \{1, 4, 7, 8\}$, $B = \{4, 8, 7, 1\}$, $C = \{8, 7, 4\}$. Then $A = B$ but $A \neq C$, $B \neq C$

Note that the order in which the elements appear in the set is not important. The other important thing to note from the above example is that although set $C \neq A$, all its elements are found in A. (Similarly, all elements of C are also found in B). The term Subset is used to describe this relationship. The section below will discuss subsets.

Subset

Definition: Let A and B be any two sets. Then A is a subset of B (or A is contained in B) if each element of A is also an element of B. This is written as $A \subseteq B$ (read as 'A is a subset of

B').

Thus, $A \subseteq B$ if whenever $x \in A$ then $x \in B$. If A is not a subset of B, we shall write $A \not\subseteq B$

The **Venn diagram** in Figure 1.1 illustrates the concept of subset:

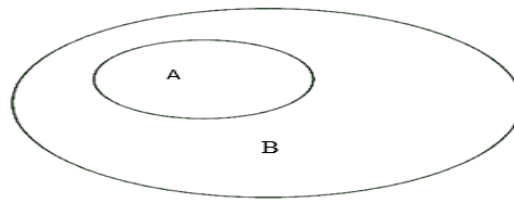


Figure 1.1: $A \subseteq B$

Notice that A consists of all elements within the smaller region while B consists of all elements within the larger region including A. Obviously, every element of A is a member of B.

Note: The concept of Venn diagram was introduced in set theory by a British mathematician, John Venn to simplify the representation of sets.

Example 1.8

Let B be the set of all Education Business Studies (EBS) students at the MUBAS, while A is the set of first year EBS students.

Since first year EBS students are Education Business Studies students too, then they are all in the set of all Education Business Studies students at the MUBAS. Thus $A \subseteq B$.

Example 1.9

Let $A = \{2, 5, 7, 9\}$, $B = \{2, 3, 4, 5, 6, 7, 8, 9\}$ and $C = \{2, 4, 6, 8\}$.

Then $A \subseteq B$ since all the elements of A are also found in B and similarly, $C \subseteq B$. On the other hand, $C \not\subseteq A$ because 4 is a member of C but not a member of A. Similarly $B \not\subseteq A$ (why is this the case?)

Proper subset: A proper subset is a special type of subset. There are two requirements for set A to be a proper subset of set B. They are:

- A is a subset of B, i.e., $A \subseteq B$

- A is not equal to B, i.e., $A \neq B$

If A is a proper subset of B, we shall denote this relationship as $A \subset B$ (read as ‘set A is a proper subset of set B’).

Example 1.10

Determine, for $A = \{\text{red}, \text{blue}\}$ and $B = \{\text{red}, \text{white}, \text{blue}\}$, whether $A \subset B$.

Solution

$A \subseteq B$ since red and blue are in both sets A and B, and $A \neq B$ since set B contains the element “white” but set A does not.

$\therefore A \subset B$ i.e. A is a proper subset of B

Empty set: So far all the sets we have looked at have at least an element in them therefore not empty or non-empty. However, it is possible to have an empty set. The **empty set** (or null set) is a set that has no members/elements.

The symbol $\emptyset$ is used to represent the empty/null set, $\{ \}$.

Example 1.11

Use set notation to describe the set of calendar months with 32 days

Solution

Using the set-builder notation, $\emptyset = \{m : m \text{ is a month with 32 days}\}$ or

Using the roster notation, let M be the set of calendar months with 32 days, $\therefore M = \{ \}$

Note: Please, always remember that

- $\{\emptyset\}$ does not symbolize the empty set; it represents a set that contains an empty set.
- $A \subseteq A$: A set is a subset of itself since a set contains all its elements.
- $\emptyset \subseteq A$: The empty set is a subset of every set because every element in the empty set

belongs to any set since the empty set has no elements.

Subsets and word problems

We can think of a subset as being ‘a selection’ from a specified group of item. Think of the group of items as a set. Then the different ways the selection can be made are the subsets.

Example 1.12

Chloe has four different colour bracelets: black (b), white (w), gold (g), and silver (s). She is deciding which ones to wear today. How many choices does she have?

Solution

Let us consider the bracelets she has to choose from as set B such that $B = \{b, w, g, s\}$. Her choices are, therefore, all possible subsets of B as follows:

Option 1: She can wear none of the bracelets

$\{ \}$ (0-element subsets – 1 choice)

Option 2: She can wear exactly 1 bracelet

$\{b\}, \{w\}, \{g\}, \{s\}$ (1-element subsets – 4 choices)

Option 3: She can wear any 2 bracelets

$\{b, w\}, \{b, g\}, \{b, s\}, \{w, g\}, \{w, s\}, \{g, s\}$ (2-element subsets – 6 choices)

Option 4: She can wear are any 3 bracelets

$\{b, w, g\}, \{b, w, s\}, \{b, g, s\}, \{w, g, s\}$ (3-element subsets – 4 choices)

Option 5: She can wear all the 4 bracelets

$\{b, w, g, s\}$ (4-element subsets – 1 choice)

$\therefore$ Chloe has $1 + 4 + 6 + 4 + 1 = 16$ different choices

Power set

Definition: Power set of a set A, denoted by $P(A)$, is the set of all subsets of the set A.

Example 1.13

Let $A = \{a, b\}$. Find the power set of A.

Solution

Subset of A are $\emptyset, \{a\}, \{b\}$ and $\{a, b\}$. $\therefore$ the power set of A is $P(A) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$



Activity 1b

1. State which of the following sets are equal:

$$A = \{a, c, t\}, B = \{t, a, b\}, C = \{c, a, t\}, D = \{a, c, t, o, r\}, E = \{t, a, c\}$$

2. List all the subsets of $B = \{10, 15, 17\}$

3. Explain whether the statement $A \subset A$ is true or false.

4. Consider the sets: $R = \{2, 4, 6, 8\}$ and $S = \{2, 4, 6, 8, 10\}$. Explain why $S \neq R$.

5. Consider these three sets: $A = \text{the set of all even numbers}$, $B = \{2, 4, 6\}$ and

$$C = \{2, 3, 4, 6\}$$

State, with a reason, which set is a proper subset of the other.

6. Write the power set of $N = \{1, 2, 3\}$.

7. Let $A = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$. Which of the following is true or false?

a) $\{1, 2\} \subset A$

b) $\emptyset \in A$



Lesson 3: Union, intersection and complement

We can also study relation between two or more sets by looking at the union or intersection of

sets.

Union of sets

Definition: Let A and B be two sets. Then the union of A and B , denoted $A \cup B$, is the set of elements that are found in either set i.e. in A only or B only or both A and B . In other words, the union consists of all elements in the two sets including those that are common to them.

In short and more formally, $A \cup B = \{x : x \in A \text{ or } x \in B\}$

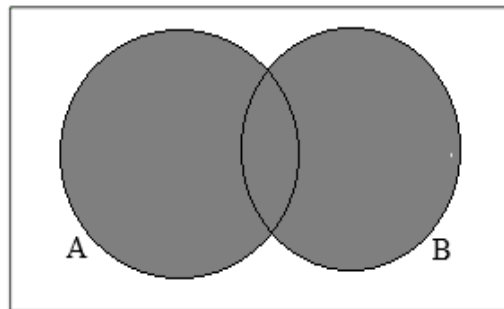


Figure 1.2: $A \cup B$
(i.e. the shaded region is the union)

We will now look at some properties of union of sets.

Proposition 1.1: Unions

Let A , B , C be any sets, then

- $A \cup A = A$: The union of a set and itself is the set itself
- $A \cup B = B \cup A$: The union of A and B is the same as the union of B and A
- $A \cup (B \cup C) = (A \cup B) \cup C$: The union of A and the union of B and C is the same as the union of the union A and B , and C
- $A \subseteq A \cup B$ and $B \subseteq A \cup B$: A is a subset of the union of A and B , and B is a subset of the union of A and B .

Intersection of sets

Definition: The intersection of two sets A and B , denoted as $A \cap B$, is defined as the set of elements that are found in both A and B i.e. elements that are common to both sets. Figure 1.3 depicts the intersection between A and B .

In short and more formally, $A \cap B = \{x : x \in A \text{ and } x \in B\}$

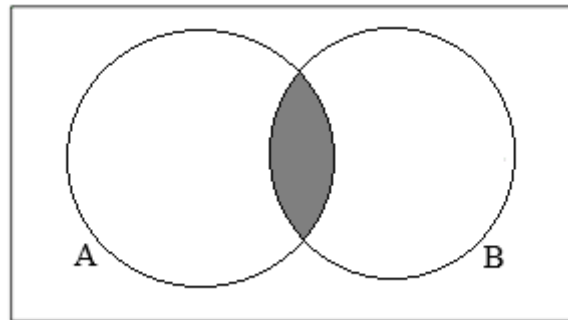


Figure 1.3: $A \cap B$
(i.e. the shaded region is the intersection)

There are times when sets do not share any common elements therefore the intersection, $A \cap B$, is a null set i.e. $A \cap B = \emptyset$. Sets that do not have common elements between them are known as **disjoint sets**.

Disjoint sets: If $A \cap B = \emptyset$, then the sets A and B are said to be **disjoint** as you can see in Figure 1.4 below.

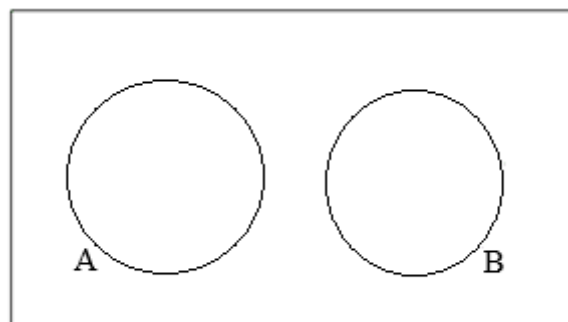


Figure 1.4: Disjoint sets
(no common elements)

Now let's consider some properties of intersections that you will find helpful as you handle sets.

Proposition 1.2: Intersections

Let A, B, C be any sets, then

- $A \cap A = A$: The intersection of a set and itself is the set itself
- $A \cap B = B \cap A$: The intersection of A and B is the same as the intersection of B and A

- $A \cap (B \cap C) = (A \cap B) \cap C$: The intersection of A and the intersection of B and C is the same as the intersection of the intersection of A and B, and C
- $A \cap B \subseteq A$ and $A \cap B \subseteq B$: The intersection of A and B is a subset of A, and the intersection of A and B is a subset of B.

Now, we can look at some examples:

Example 1.14

Find $A \cup B$ if $A = \{1, 2, 4, 7\}$ and $B = \{2, 7, 9, 11, 12\}$.

Solution

$$A \cup B = \{1, 2, 4, 7, 9, 11, 12\}$$

Note that we do not repeat the 2's and the 7's.

Example 1.15

Supposed $A = \{3, 6, 9\}$, $B = \{3, 4, 5, 6, 7\}$, and $C = \{4, 5, 7\}$. Find

- $A \cap B$
- $A \cap C$

Solution

a) Elements found in both A and B are 3 and 6, $\therefore A \cap B = \{3, 6\}$

b) Since there are no elements that are found in both A and C, $\therefore A \cap C = \emptyset$ or $\therefore A \cap C = \{ \}$

Complement of a set

In order to understand the complement of a set, you must first of all learn what a universal set is. A **universal set**, denoted U , is the set of all elements under consideration. This is usually

defined by the context. For instance, if $\{ : A \text{ is a student in EBS1} \}$,

$B = \{ b \mid b \text{ is a student in EBS} \}$, and $C = \{ c \mid c \text{ is a student in EBS} \}$, then the universal set would be defined as the set of all EBS students i.e. $U = \{ x : x \text{ is an EBS student} \}$.

It is important to mention that you may also come across the symbol, ξ for a universal set (ξ is the Greek letter xi and *pronounced Shi or Ksi*). However, note that we will always use U to denote the universal set in this module.

Definition: Complement

Let U be the universal set and A be any set such that $A \subseteq U$, then the **complement** of A in U , denoted A^c (read as ‘ A complement’) is the set of all elements that are not in A but in U . The Venn diagram in Figure 1.5 shows the complement of set A .

In short and more formally, $A^c = \{ x \in U : x \notin A \}$

Note: You can also denote a complement of set A as A' or $\bar{A}$ (Both of these are acceptable).

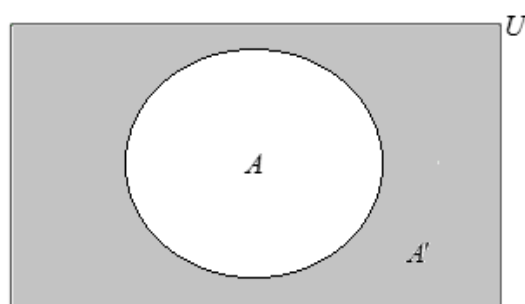


Figure 1.5: A^c
(i.e. the shaded region is the complement of A)

Let's consider a few examples to consolidate the concepts of universal set and complement.

Difference of sets

Let's consider any two sets, A and B . The difference of set A and B , denoted $A - B$, is the collection of all elements which are in A but not in B . Similarly, the difference of B and A denoted, $B - A$, is the collection of all elements which are in B but not in A .

In general, if A and B are two sets, then:

$$A - B = \{x : x \in A \text{ and } x \notin B\} \text{ and } B - A = \{x : x \in B \text{ and } x \notin A\}$$

The difference of two sets can be represented using Venn diagram as shown in Figure 1.6.

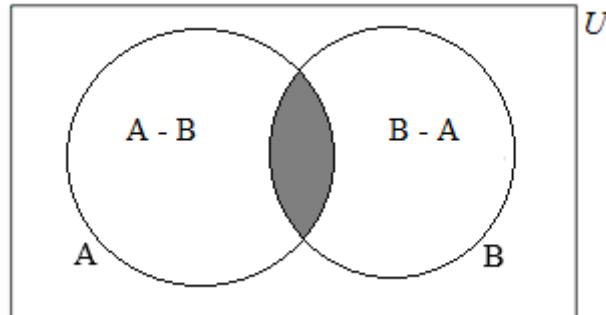


Figure 1.6: Difference of sets
(unshaded areas within the circles)

Example 1.16

Let the universal set, $U = \{a, b, c, w, z, 1, 2, 3, 4, 5\}$, $A = \{1, 2, w, z\}$ and $B = \{2, 4, 5, a, b, w\}$. Find the following:

- A'
- B'

Solution

- $A' = \{a, b, c, 3, 4, 5\}$
- $B' = \{1, 3, c, z\}$

Example 1.17

Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{2, 4, 6\}$. Find

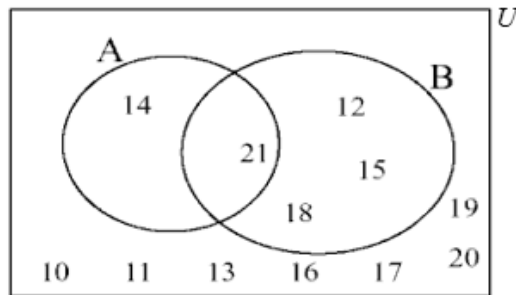
- $A - B$
- $B - A$

Solution

- $A - B = \{1, 3, 5\}$ i.e. elements which are in A but not in B
- $B - A = \{6\}$ i.e. elements which are in B but not in A

Example 1.18

Consider the following Venn diagram below:



Represent the following using roster notation:

- | | | |
|-------------------|---------------|-----------------|
| a) U | d) $A \cup B$ | g) $A' \cap B'$ |
| b) $A \cap B$ | e) A^c | h) $A' \cap B$ |
| c) $(A \cap B)^c$ | f) B^c | i) $A - B$ |

Solution

- a) $U = \{10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21\}$
- b) $A \cap B = \{21\}$
- c) $(A \cap B)^c = \{10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20\}$
- d) $A \cup B = \{12, 14, 15, 18, 21\}$
- e) $A^c = \{10, 11, 12, 13, 15, 16, 17, 18, 19, 20\}$
- f) $B^c = \{10, 11, 13, 14, 16, 17, 19, 20\}$
- g) $A' \cap B' = \{10, 11, 13, 16, 17, 19, 20\}$
- h) $A' \cap B = \{12, 15, 18\}$
- i) $A - B = \{14\}$

You might have noticed from the two examples above that $A \subseteq U$ and $B \subseteq U$ as discussed

earlier on. Using this fact, we can state some propositions or mathematical statements about the universal set that are true any set A.

Proposition 1.3: Set A in U

Let U be the universal and A be any set such that $A \subseteq U$, then

- $(A')' = A$: The complement of a complement of A is the set A itself
- $A \cup U = U$: The union of A and the universal set is the universal set
- $A \cap U = A$: The intersection of A and the universal set is the set A itself
- $A \cup \emptyset = A$: The union of A and an a null set is the set A itself
- $A \cap \emptyset = \emptyset$: The intersection of A and a null set is a null set
- $A \cup A' = U$: The union of A and its complement is the universal set
- $A \cap A' = \emptyset$: The intersection of A and its complement is a null set
- $U - A = A^c$: The difference of the universal set and A is the complement of A
- $U' = \emptyset$: The complement of the universal set is a null set
- $\emptyset' = U$: The complement of a null set is the universal set

Cartesian product of two sets

Definition: We will define the Cartesian product of two sets as the set of all ordered pairs of elements of the two sets.

In general if $A = \{a_i : i = 1, 2, 3, 4, 5, 6 \dots n\}$ and $B = \{b_j : j = 1, 2, 3, 4, 5, 6 \dots, n\}$, the Cartesian product of A and B is

$$A \times B = \{a_i b_j : i = 1, 2, 3, 4, 5, 6 \dots, n \text{ \& } j = 1, 2, 3, 4, 5, 6 \dots, n\} \text{ while}$$

$$B \times A = \{(b_j, a_i) : i = 1, 2, 3, 4, 5, 6 \dots, n \text{ \& } j = 1, 2, 3, 4, 5, 6 \dots, n\}$$

Alternatively, we can state the same as:

$$A \times B = \{(a, b) : a \in A \text{ and } b \in B\}_{\text{while}}$$

$$B \times A = \{(b, a) : b \in B \text{ and } a \in A\}$$

Note that $A \times B \neq B \times A$ and if $A = \emptyset$ or $B = \emptyset$ then $A \times B = B \times A = \emptyset$

Example 1.19

If $A = \{1, 2\}$ and $B = \{3, 4, 5\}$, find

- a) $A \times B$
- b) $B \times A$

Solution

- a) To find the Cartesian product $A \times B$, we must find the ordered pairs i.e. all possible pairs of (a, b) . The pairs are $(1, 3), (1, 4), (1, 5), (2, 3), (2, 4), (2, 5)$

$$\therefore A \times B = \{(1, 3), (1, 4), (1, 5), (2, 3), (2, 4), (2, 5)\}$$

- b) To find Cartesian product $B \times A$, we must find the ordered pairs i.e. all possible pairs of (b, a) . The pairs are $(3, 1), (3, 2), (4, 1), (4, 2), (5, 1), (5, 2)$

$$\therefore B \times A = \{(3, 1), (3, 2), (4, 1), (4, 2), (5, 1), (5, 2)\}$$

Let's finish this unit with a word problem example that will summarize what we have learnt so far and show how to apply set theory to solving real life problems.

Example 1.20

From a survey of 100 students at Mtendere Secondary School, it was found out that 75 students participate in environmental projects, 45 students participate in orphan care activities and 35 participate in environmental projects and orphan care activities.

- a) How many students participate in either environmental projects or orphan care activities?
- b) How many students did not participate in either environmental projects or orphan care activities?

Solution

Venn diagrams are very useful tools for solving this type of problems and therefore we shall

use a Venn diagram to find the solution to the problem. If we let

U = set of all students in the sample, $n(U) = 100$

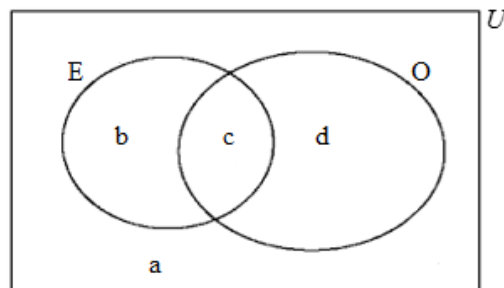
E = set of all students who participate in environmental projects, $n(E) = 75$

O = set of all students who participate in orphan care activities, $n(O) = 45$

$E \cap O$ = set of all students who participate in environmental projects and orphan care activities, $n(E \cap O) = 35$

We will use small letters to represent numbers of students who participate or not participate in the respective activities. Thus in the Venn diagram below: a represents those that do not participate in any of the activities, b represents those that participate in environmental projects only, c represents those that participate in both environmental projects and orphan care activities while d represents those that participate in orphan care activities only.

Then our Venn diagram looks like this:



We now form basic equations as follows:

Universal set consists of all the 100 students i.e. $n(U) = 100$

$$a + b + c + d = 100 \dots\dots\dots (i)$$

Participating in environmental projects i.e. $n(E) = 75$

$$b + c = 75 \dots\dots\dots (ii)$$

Participating in orphan care activities i.e. $n(O) = 45$

$$c + d = 45 \dots\dots\dots (iii)$$

Participating in both i.e. $n(E \cap O) = 35$, $\therefore c = 35$

Since $c = 35$, then from (ii), $b + c = 75$, $\therefore b = 75 - c = 75 - 35 = 40$

From (iii), $c + d = 45$, $\therefore d = 45 - c = 45 - 35 = 10$

and from (i), $a + b + c + d = 100$, $\therefore a = 100 - b - c - d = 100 - 40 - 35 - 10 = 15$

Hence,

- a) Those that participate in environmental or orphan care activities are:

$$n(E \cup O) = b + c + d = 40 + 35 + 10 = 85$$

85 students participate in environmental or orphan care activities

- b) Those that did participate in any of the activities are:

$$n(E \cup O)' = a = 15$$

15 students did not participate in either environmental projects or orphan care activities.

Remarks:

In Example 1.18, we have seen how powerful Venn diagrams are as a tool for solving real-life problems that arise out of the concept of set. Notice also how the lower case letters have been used to represent numbers for students participating in particular activities and then eventually forming equations. You are strongly encouraged to make use of Venn diagrams in solving problems whenever necessary.

Summary

In this unit, you learnt that:

- A **set** is a collection of distinct objects, called elements or members of the set.
- A set can be defined by describing the contents or by listing the elements of the set, enclosed in curly brackets.
- A set can be described using either the **set builder notation** or **roster notation**.
- A **subset** is another set that contains only elements from the original set
- A **proper subset** is a subset that is not identical to the original set - contains fewer elements
- A set that contains no elements is called an **empty set** and is notated $\emptyset$ or $\{ \}$
- The union of two sets contains all the elements contained in either set (or both sets)
- The intersection of two sets contains only the elements that are in both sets.
- A universal set is a set that contains all the elements we are interested in and is

contextual.

- A complement of a set is relative to the universal set, and it contains all the elements in the universal set that are not in the set.
- Power set is the set of all subsets of a given set
- Venn diagram represents each set by a circle, usually drawn inside of a containing box representing the universal set. Overlapping areas indicate elements common to both sets.

The next unit is about Number Systems. The unit will afford you an opportunity to appreciate how sets are used in classifying numbers.



Further reading

You are encouraged to read the following before attempting the unit exercises. E-copies (pdfs) of these reference materials may be available. Please request them if they are not made available.

Ewer, J. P. G. (1994). Algebra, trigonometry and calculus. Zomba: Science

Teachers Association of Malawi. **(Pages: 1 to 12)**

Lial, M. L., Hornsby, J. & Schneider, D. I. (2017). *College algebra and trigonometry* (6th ed.). New York: Pearson. **(Section R.1 & R.2)**

Swokowski, E. W., & Cole, J. (2012). *Algebra and trigonometry with analytical geometry* (13th ed.). Belmont: Brooks/Cole. **(1 to 15, 87 to 93)**

Ummer, E. K. (2012). Basic mathematics for economics, business, and finance. London: Routledge. **(Pages: 2 to 6)**

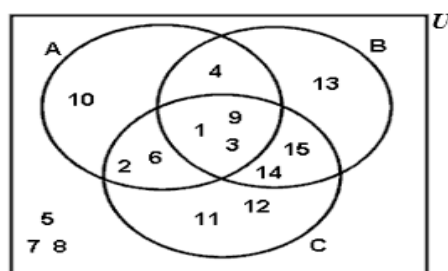
Website/s: <http://mathworld.wolfram.com/SetTheory.html>



Unit exercises

1. Use the set builder notation to represent the set of all real numbers such that $x^2 = 9$.
2. Given that $M = \{0, 1, 3, 7\}$, $N = \{0, 2, 3, 8\}$ and $P = \{-2, 0, 2, 3, 9\}$
 - a) Show the three sets on a Venn diagram.
 - b) What is cardinality of the intersection of the three sets i.e. $n(M \cap N \cap P)$?
3. Given that X, Y and Z are sets with some common elements.

- a) Draw a Venn diagram and shade the region representing $X' \cap Y' \cap Z'$
- b) Find $n(X' \cap Y' \cap Z')$ if $n(X \cup Y) = 11$ and $n(X \cup Y \cup Z) = 14$.
4. At a certain school, there are 200 students. Every student studies at least one of the subjects: English, Mathematics and Biology. 15 students study all the three, 28 study English and Biology, 30 students study English and Mathematics, 58 study Mathematics and Biology, 62 study English and 25 study Mathematics only. How many students study Biology only?
5. A survey asks 200 people “What beverage do you drink in the morning”, and offers three choices: Tea only, coffee only, and both coffee and tea. Suppose 20 respond tea only, 80 respond coffee only, 40 respond both. How many people drink
- a) tea in the morning?
- b) neither tea nor coffee?
6. Let $A = \{a, b, c, d\}$, $B = \{e, f\}$ and $C = \{23, 35\}$. Find
- a) $P(A)$
- b) $A \times B$
- c) Cardinality of $C \times B$
- d) $B - C$
7. Consider the Venn diagram below



Described the following using set roster notation:

- a) B^c
- b) $(A \cup B) \cap C^c$
- c) $(A \cap B \cap C)$

